SQL PROJECT

DATA BANK

by

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Introduction

Data Bank is a digital only platform that combines banking services with secure cloud data storage, where customers storage limits depend on their account balances.

As the company aims to grow, it needs to analyze customer behavior, track key metrics, and forecast storage demand. This case study focuses on using data analytics to generate insights that support efficient planning, resource optimization, and business growth.

Problem Statement

Data Bank needs to analyze customer data and financial activity to accurately forecast data storage demand while supporting customer growth and optimizing resource allocation.

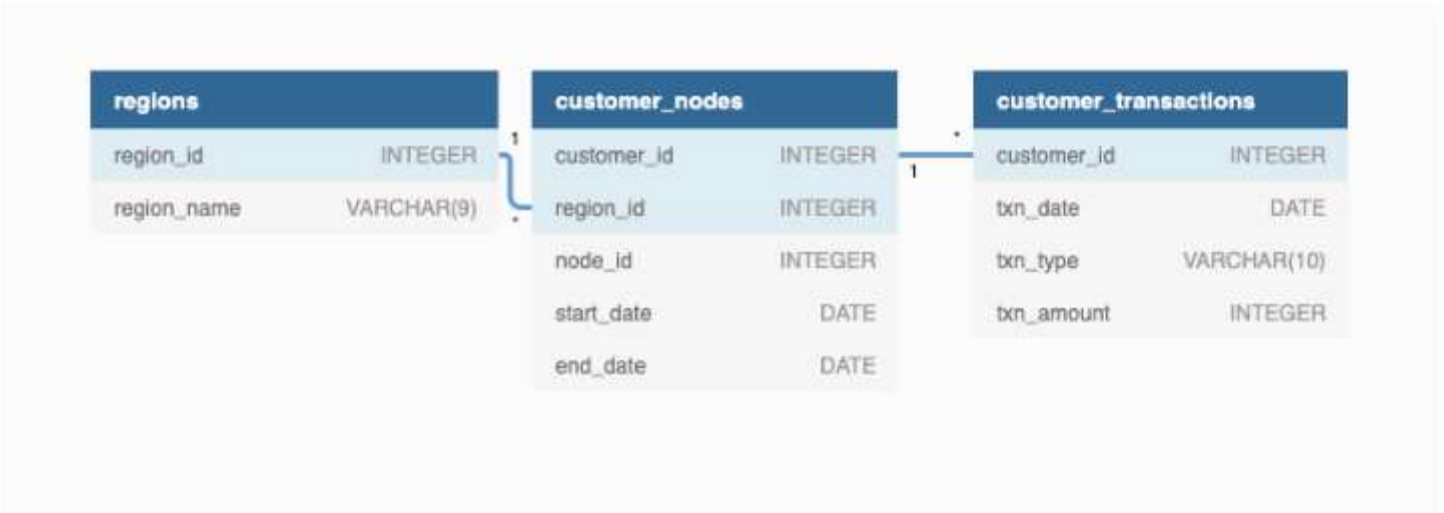
Entity Relationship Diagram

The 3 key datasets for this case study

Regions

Customer nodes

Customer transactions



Key Business Questions

- How is the customer base growing?
- What are the transaction patterns?
- How much data storage is required per customer?
- What will future storage demand look like?

```sql
--- Business Questions

--- 1. How many unique nodes are there on the Data Bank system?

SELECT COUNT(DISTINCT node_id) AS unique_nodes
FROM customer_nodes;
```

100 % No issues found

Results | Messages

	unique_nodes
1	5

```sql
--- 2. What is the number of nodes per region?

SELECT
    region_id,
    COUNT(DISTINCT node_id) AS nodes_per_region
FROM customer_nodes
GROUP BY region_id;
```

100 % No issues found

Results Messages

	region_id	nodes_per_region
1	1	5
2	2	5
3	3	5
4	4	5
5	5	5

```sql
--- 3. How many customers are allocated to each region?

SELECT
    region_id,
    COUNT(DISTINCT customer_id) AS customers
FROM customer_nodes
GROUP BY region_id;
```

100 % No issues found

Results Messages

	region_id	customers
1	1	47
2	2	43
3	3	33
4	4	41
5	5	36

```sql
--- 4. How many days on average are customers reallocated to a different node?

SELECT
    AVG(DATEDIFF(DAY, start_date, end_date)) AS avg_reallocation_days
FROM customer_nodes
WHERE end_date IS NOT NULL;
```

100 % No issues found

Results Messages

	avg_reallocation_days
1	1457269

Data_bank A...\Doxxa (68))*

```sql
--- 5. What is the median, 80th and 95th percentile for this same reallocation days metric for each region?

WITH reallocation AS (
    SELECT
        region_id,
        DATEDIFF(DAY, start_date, end_date) AS days_diff
    FROM customer_nodes
    WHERE end_date IS NOT NULL
)
SELECT DISTINCT
    region_id,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY days_diff)
        OVER (PARTITION BY region_id) AS median_days,
    PERCENTILE_CONT(0.8) WITHIN GROUP (ORDER BY days_diff)
        OVER (PARTITION BY region_id) AS p80_days,
    PERCENTILE_CONT(0.95) WITHIN GROUP (ORDER BY days_diff)
        OVER (PARTITION BY region_id) AS p95_days
FROM reallocation;
```

100 % No issues found

Results | Messages

	region_id	median_days	p80_days	p95_days
1	1	1457240	2914526.4	2914559.05
2	2	1457249.5	2914526	2914547.25
3	3	1457257	2914532	2914555.75
4	4	1457249	2914534	2914552.95
5	5	1457253	2914524	2914546.95

```sql
--- 6. What is the unique count and total amount for each transaction type?

SELECT
    txn_type,
    COUNT(*) AS transaction_count,
    SUM(txn_amount) AS total_amount
FROM customer_transactions
GROUP BY txn_type;
```

100 % No issues found

Results Messages

	txn_type	transaction_count	total_amount
1	deposit	118	55806
2	purchase	82	38436
3	withdrawal	88	43448

```sql
--- 7. What is the average total historical deposit counts and amounts for all customers?

WITH customer_deposits AS (
    SELECT
        customer_id,
        COUNT(*) AS deposit_count,
        SUM(txn_amount) AS total_amount
    FROM customer_transactions
    WHERE txn_type = 'deposit'
    GROUP BY customer_id
)
SELECT
    AVG(deposit_count) AS avg_deposit_count,
    AVG(total_amount) AS avg_deposit_amount
FROM customer_deposits;
```

100 % No issues found

Results | Messages

	txn_type	transaction_count	total_amount
1	deposit	118	55806
2	purchase	82	38436
3	withdrawal	88	43448

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```sql
--- 8. For each month how many Data Bank customers make more than 1 deposit and either 1 purchase or 1 withdrawal in a single month?

WITH monthly_txns AS (
    SELECT
        customer_id,
        FORMAT(txn_date, 'yyyy-MM') AS month,
        SUM(CASE WHEN txn_type = 'deposit' THEN 1 ELSE 0 END) AS deposits,
        SUM(CASE WHEN txn_type = 'purchase' THEN 1 ELSE 0 END) AS purchases,
        SUM(CASE WHEN txn_type = 'withdrawal' THEN 1 ELSE 0 END) AS withdrawals
    FROM customer_transactions
    GROUP BY customer_id, FORMAT(txn_date, 'yyyy-MM')
)
SELECT
    month,
    COUNT(DISTINCT customer_id) AS customers
FROM monthly_txns
WHERE deposits > 1
  AND (purchases >= 1 OR withdrawals >= 1)
GROUP BY month;
```

100 % No issues found Ln: 837, Ch: 45 SPC

Results | Messages

	month	customers
1	2020-01	3
2	2020-02	10
3	2020-03	8
4	2020-04	4

Data_bank A...\Doxxa (68))*

```sql
--- 9. What is the closing balance for each customer at the end of the month?


WITH running_balance AS (
    SELECT
        customer_id,
        txn_date,
        DATEFROMPARTS(YEAR(txn_date), MONTH(txn_date), 1) AS month_start,
        SUM(txn_amount) OVER (
            PARTITION BY customer_id
            ORDER BY txn_date
            ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
        ) AS closing_balance
    FROM customer_transactions
),

month_end_balance AS (
    SELECT
```

100 % No issues found

Results | Messages

	customer_id	month_start	closing_balance
1	1	2020-03-01	1600
2	8	2020-01-01	1512
3	8	2020-02-01	1692
4	8	2020-03-01	2934
5	8	2020-04-01	3906
6	27	2020-01-01	2325
7	27	2020-02-01	5831
8	27	2020-03-01	8407
9	29	2020-01-01	1409
10	29	2020-02-01	3221
11	29	2020-03-01	6092
12	29	2020-04-01	9071

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```sql
--- 10. What is the percentage of customers who increase their closing balance by more than 5%?


WITH monthly_balance AS (
    SELECT
        customer_id,
        DATEFROMPARTS(YEAR(txn_date), MONTH(txn_date), 1) AS month_start,
        SUM(txn_amount) OVER (
            PARTITION BY customer_id
            ORDER BY txn_date
        ) AS closing_balance
    FROM customer_transactions
),

balance_with_prev AS (
    SELECT
        customer_id,
        month_start,
```

100 % No issues found

Results Messages

	percentage_of_customers
1	96.550000000000

Key Insights

- Customer balances directly impact storage allocation
- Growth trends influence infrastructure demand
- Usage patterns vary across customer segments

Recommendations

- Implement scalable storage solutions
- Monitor customer growth trends continuously
- Optimize resource allocation based on usage patterns

THANK YOU!